

Homework 1 — Programs that require calculations

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Due Sunday 27 September, 23:59, on Canvas. The deadline is fixed; late work is not received.

Individual work. Two programs.

What you submit

For **each** program, three things:

1. **The handwritten algorithm** — photograph or scan of paper, or the A_Matricula.docx template. INPUT: line, OUTPUT: line, numbered steps.
2. **The .py file**, named as stated. It starts the way every program in this course starts — and the way the quick exam expects: **line 1, your name in a # comment; then the algorithm inside a """ multi-line comment; then the code.**
3. **Your check** — one line stating the test values you used and the result you got.

Run at least one of the two programs in **Trinket** (<https://trinket.strivemath.org/python3>) as well as Thonny. It is the editor of Tuesday's quick exam.

A submission with code but no algorithm is **incomplete** and is graded as such. The algorithm is the work; the code is the transcription. See *Using AI in this course*.

Problem 1 — Body Mass Index

$BMI = \text{weight in kilograms} \div (\text{height in metres})^2$.

Write a program that reads a person's weight and height and displays their BMI, rounded to one decimal place.

Save as: T1_A_matricula.py

Required check: 70 kg, 1.75 m → 22.9

Precedence matters here. `weight / height ** 2` and `weight / height * height` are not the same program, and only one of them is right. Be able to say which and why.

Problem 2 — Seconds to time

Write a program that reads a whole number of seconds and displays the equivalent time in hours, minutes and seconds.

Save as: T1_B_matricula.py

Required checks:

- 3671 → 1 h, 1 min, 11 s
- 59 → 0 h, 0 min, 59 s
- 7200 → 2 h, 0 min, 0 s

You need two operators from Friday that you may not have used yet: `//` (whole-number division) and `%` (remainder). 3671 `//` 3600 is 1. 3671 `%` 3600 is 71. Work out the rest on paper — that is the exercise.

Use `int()` here, not `float()`. Nobody wants 1.0 hours and 11.0 seconds.

How this is graded

Algorithm present, with input/output and numbered steps	40%
Program runs and produces correct results for the stated checks	40%
Correct use of <code>int()</code> / <code>float()</code> , sensible variable names, output readable	20%

Sensible variable names means `height_m`, not `h` and not `x`. You will read this code again in November.

If you get stuck

Asesorías, or ask in class on Tuesday — the session opens with questions. What you may **not** do is have an assistant write it: this is exactly the homework that the Tuesday quick exam tests, and the quick exam is closed-environment. Getting this wrong now, by yourself, is worth more than getting it right with help.